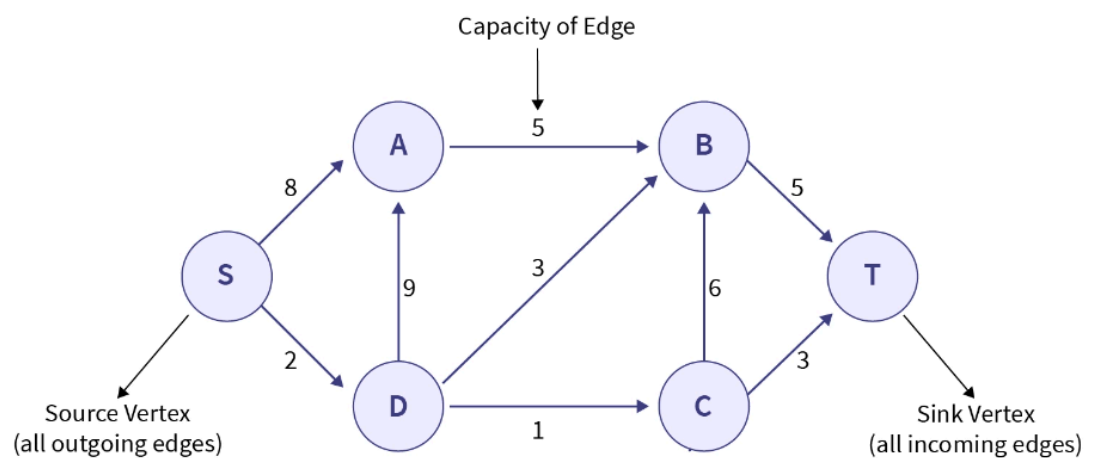


Maximum Flow Problem

In Graph Theory, **maximum flow** is the maximum amount of flow possible from source node to sink node in a given flow network.

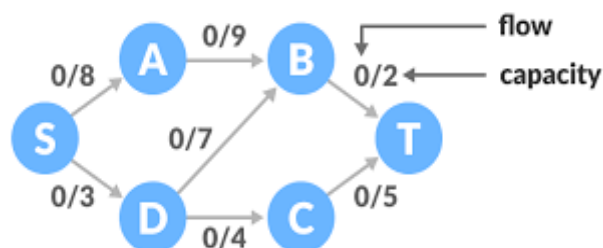
- Flow network

A **flow network**, also known as a network flow or simply a flow graph, is a directed graph where each edge has a capacity and represents a flow between two vertices (nodes).



A **flow network** should satisfy the following conditions :

- ❖ For all the vertices (except the source and the sink vertex), input flow must be equal to output flow.
- ❖ we can not send more flow through an edge than its capacity.
- ❖ Total outflow from the source vertex must be equal to total inflow to the sink vertex.



Ford Fulkerson method

The **Ford-Fulkerson algorithm** is used to detect maximum flow from start vertex to sink vertex in a given graph. In this graph, every edge has the capacity.

- **Residual Capacity** - Residual capacity of the directed edge is nothing but capacity of the edge minus current flow through the edge.
- **Augmenting Path** - It is a simple path from source to sink in a flow network. It is a route from the source to the destination in a flow network along which you can increase the flow. This path is constructed based on the residual capacities of the edges in the network.

STEPS

1.Initialize: Set the flow of all edges to 0.

2.Find an augmenting path: Path from source node to sink node. This path should have available capacity, meaning you can increase the flow along it.

3.Calculate the bottleneck capacity: Determine the maximum amount of flow that can be pushed through the augmenting path found in step 2. This is called the bottleneck capacity and it is equal to the minimum residual capacity of all edges in the augmenting path.

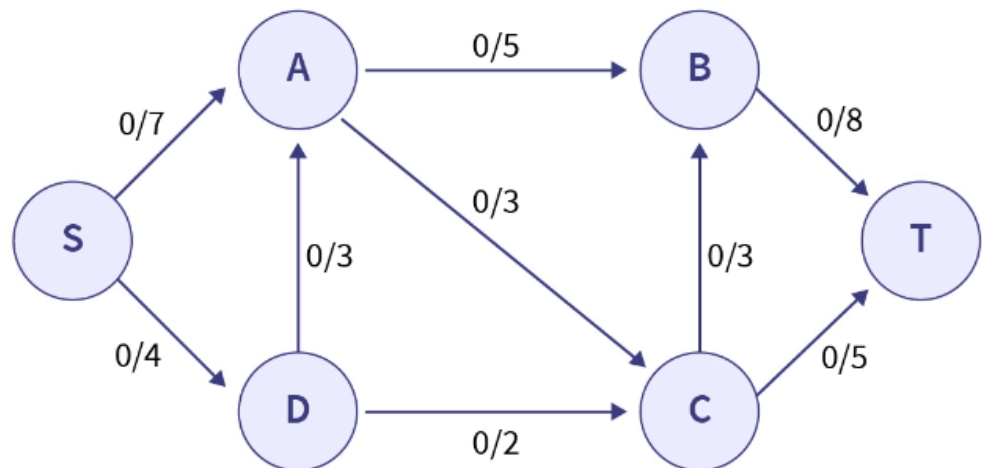
4.Update the flow: This involves adding the bottleneck capacity to the flow of each edge in the augmenting path.

5.Update the residual capacities: Update the residual (remaining) capacities of the edges in the flow network based on the flow added in step 4.

6.Repeat: Repeat steps 2-5 until no augmenting path can be found in the residual graph. Once there are no more augmenting paths, maximal flow is achieved.

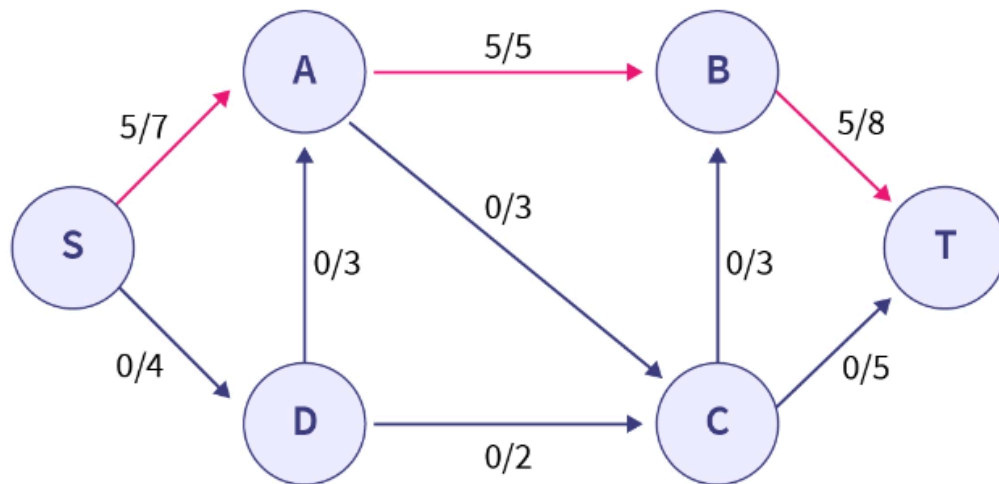
Example

Consider the following flow network.

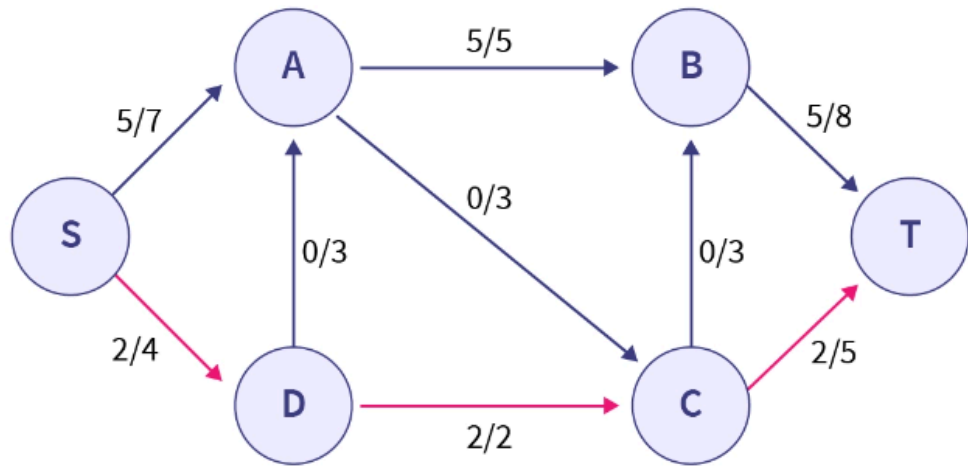


We can see that the initial flow of all the paths is 0. Now we will search for an augmenting path in the network.

- ★ One such path is **S**→**A**→**B**→**T** with residual capacities **7, 5, 8**. Their minimum is **5**, so as per the Ford-Fulkerson method we will increase a flow of **5** along the path. Then, we got

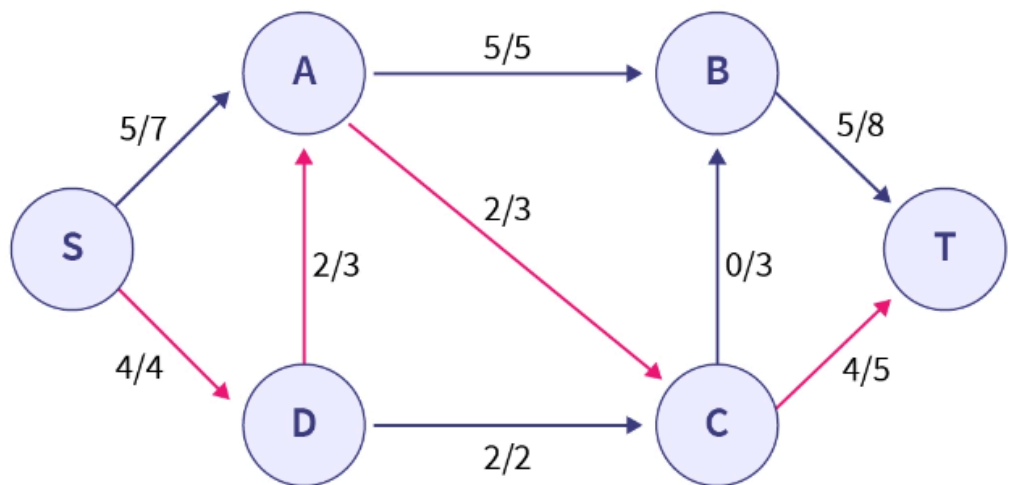


- ★ Now, we will check for other possible augmenting paths, one such path can be **S**→**D**→**C**→**T** with residual capacities as **4, 2, and 5** of which **2** is the minimum. So we will increase a flow of 2 along the path.

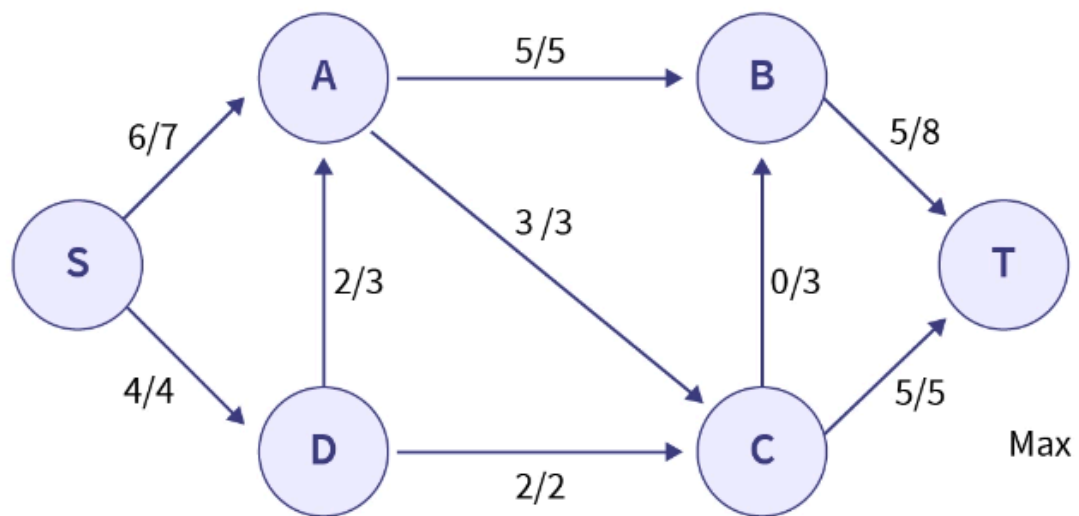


we can't increase the flow along paths **A→B** and **D→C** as their flow has reached their maximum capacity. So to find other augmenting paths, we must take care that these edges must not be part of the path.

- ★ Another path is **S→D→A→C→T** with capacities 2,3,3 and 3. 2 is the minimum, so increase a flow of 2 along the path.



- ★ Next augmenting path possible is **S→A→C→T** with capacities 2,1,1, where 1 is minimum. So increase flow of 1 along the path. after which our network will look like -



Now there are no more augmenting paths possible in the flow network, so now we can find maximum flow.

Maximum Flow = Total flow going out of **source** (flow of edge **S-A** + flow of edge **S-D**) = **6 + 4 = 10**

OR

Maximum Flow = Total flow coming in **sink** (flow of edge **B-T** + flow of edge **C-T**) = **5 + 5 = 10**